

TIC 342586820

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS

Period 1.951130 d
Epoch (t0) 1597.7365 BTJD
Duration 0.1700 d
Depth 2.926 % (29,261 ppm)
SNR 196.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

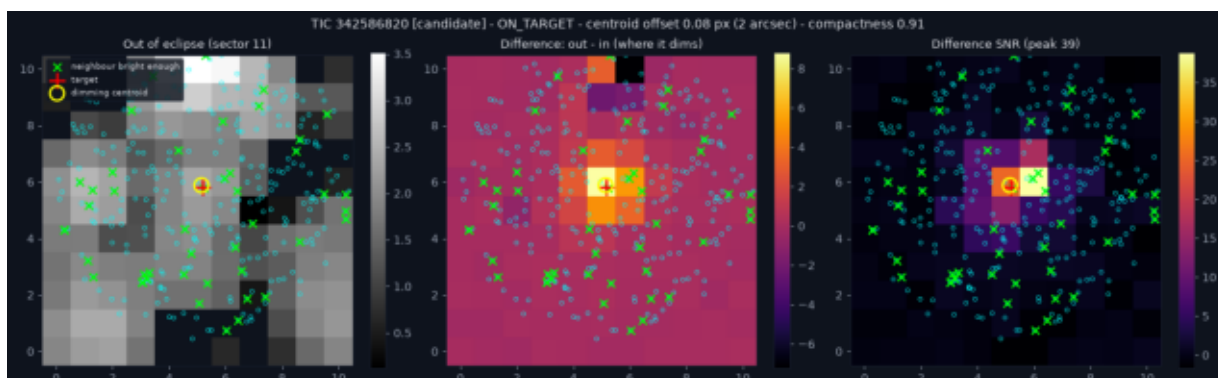
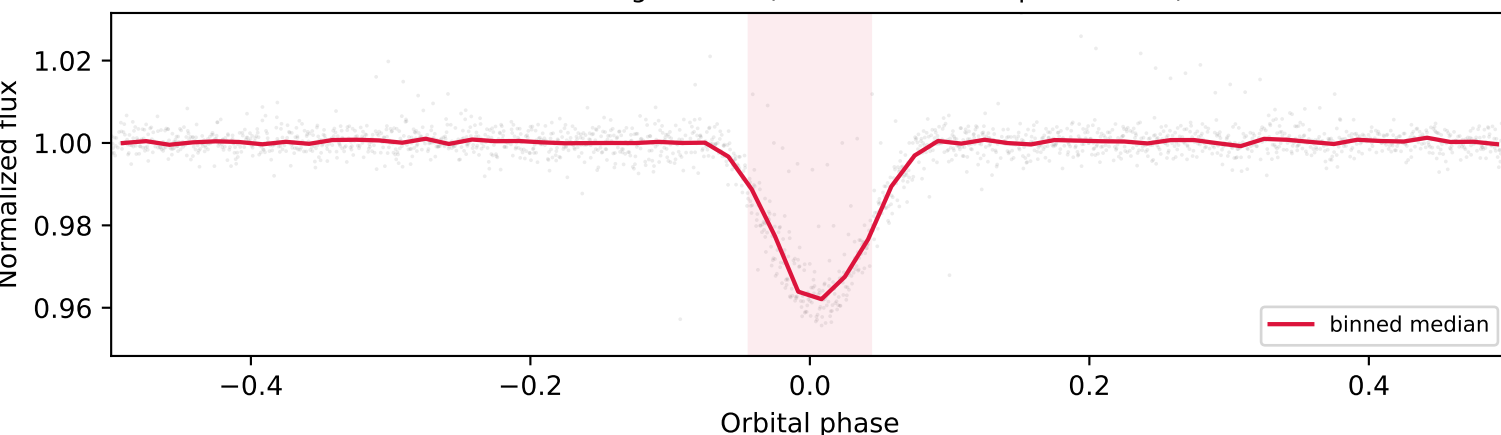
Eclipses observed 28 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 141.9 (gate ≥ 7.0)
Odd/even z-score 0.32 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 120.9 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 38.91 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.08 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 1006143852
0.14 px away, dTmag +1.80 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.069 px (dimming centroid 0.075 px from target vs 0.144 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.